

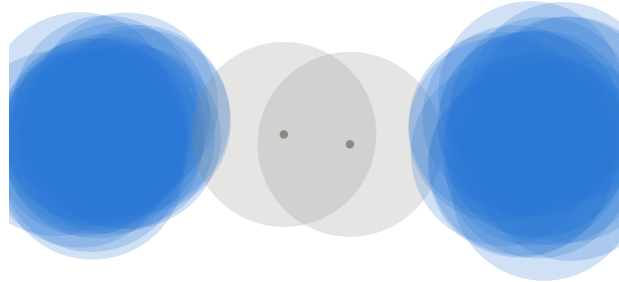
# Same cloud, same intermediate value $r = 1.3$

initial point cloud ( $r = 0$ )



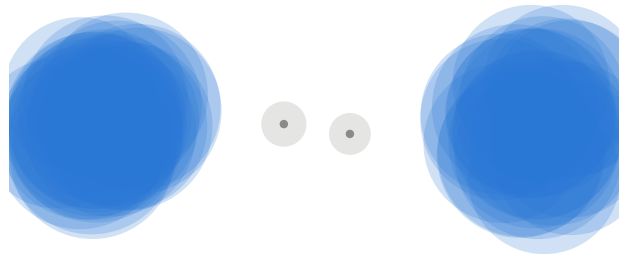
• blob • bridge

Rips, balls of radius  $r/2$   
(1 component at this  $r$ )

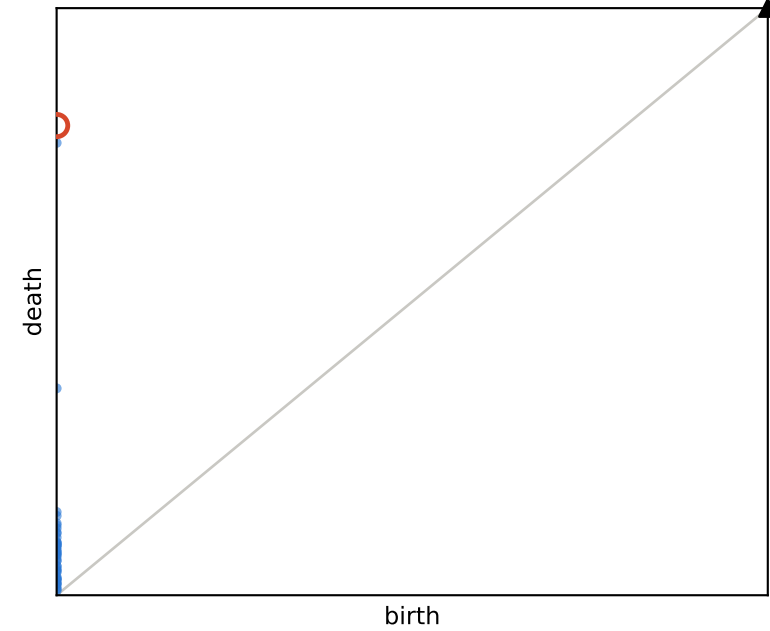


Two dense blobs joined by a sparse two-point bridge: the shortest path between the blobs, but through a neighbourhood far sparser than either blob's interior.

DTM<sub>5</sub>, balls of radius  $(r - w_i) + 1/2$   
(4 components at this  $r$ )



Rips  $H_0$



DTM<sub>5</sub>  $H_0$

